

Brain-Computer Interface Based Smart Vehicle

PROJECT-21ECP302L

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BONAFIDE CERTIFICATE

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ABSTRACT

The growing incidence of physical disabilities and neurological disorders necessitates advanced assistive technologies to restore mobility and independence. This project presents a Brain-Computer Interface (BCI)-based smart vehicle that enables hands-free control by interpreting Electroencephalography (EEG) and Electro-Oculography (EOG) signals. Bio-signals are acquired using scalp electrodes and processed via a BioAmp EXG module interfaced with an Arduino Uno R4. A threshold-based algorithm with beta-wave hysteresis and a 400 ms focus-hold mechanism classifies focused, relaxed, and blink states, which are mapped to directional vehicle commands. Wireless communication is established using ESP8266 and ESP32 modules for real-time control.

The system integrates an HC-SR04 ultrasonic sensor for obstacle detection, an MPU6050 IMU for accident detection, and a Telegram Bot for emergency alerts. Experimental validation in wireless mode achieved over 90% command accuracy, 90% obstacle-stop reliability, and sub-100 ms alert latency. The results demonstrate a low-cost, scalable, and efficient mobility solution for physically challenged individuals, aligning with SDG 3: Good Health and Well-being and SDG 10: Reduced Inequalities by promoting inclusive healthcare and accessibility.



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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iv
	ACKNOWLEDGEMENT	v
1	INTRODUCTION	1
	1.1 Background	1
	1.2 Problem Statement	1
	1.3 Motivation	1
	1.4 Objectives	2
	1.5 Scope of the Work	2
2	LITERATURE SURVEY	3
3	SYSTEM DESCRIPTION AND METHODOLOGY	6
	3.1 System Overview	6
	3.2 System Block Diagram	6
	3.3 Hardware Components	7
	3.3.1 Scalp Electrodes and BioAmp EXG Module	7
	3.3.2 Arduino Uno R4 Microcontroller	7
	3.3.3 ESP8266 Wi-Fi Module (Acquisition Node)	7
	3.3.4 ESP32 Microcontroller (Vehicle Node)	7
	3.3.5 L298N Motor Driver and DC Motors	8
	3.3.6 HC-SR04 Ultrasonic Sensor	8
	3.3.7 MPU6050 IMU (Accident detection)	8
	3.3.8 Telegram Bot Integration	8
	3.4 Implementation	8
	3.4.1 signal Processing Pipeline	8
	3.4.2 Command Mapping Table	9
	3.4.3 Wireless Communication Architecture	10
	3.4.4 Safety Architecture	10
	3.4.5 PWM Speed calibration	10
4	RESULTS AND DISCUSSION	11
	4.1 Hardware Implementation	11

CHAPTER NO.	TITLE	PAGE NO.
	4.2 Signal Acquisition and processing	12
	4.3 Vehicle Control Validation	13
	4.4 Ultrasonic Sensor Validation	13
	4.5 MPU6050 Accident Detection and Telegram Alert	13
	4.6 Discussion	14
5	CONCLUSION	15
	5.1 Conclusion	15
	5.2 Further Directions	15
	REFERENCES	17

LIST OF TABLES

TABLE NO	TITLE	PAGE NO
3.1	Command Mapping	9
4.1	Vehicle Control Validation	13

LIST OF FIGURES

FIG NO	TITLE	PAGE NO
3.1	Block Diagram of the proposed System	6
4.1	Real Time System Implementation	11
4.2	Real Time Signal Acquisition	12

LIST OF ABBREVIATIONS

SNO	ABBREVIATION	DESCRIPTION
1	ADC	Analog to Digital Converter
2	ALS	Amyotrophic Lateral Sclerosis
3	AP	Access Point
4	ARM	Advanced RISC Machine
5	BCI	Brain-Computer Interface
6	CMRR	Common Mode Rejection Ratio
7	EEG	Electroencephalography
8	EOG	Electro-Oculography
9	ESP	Espressif Systems Platform
10	FIR	Finite Impulse Response

11	GPIO	General Purpose Input/Output
12	HTTPS	Hypertext Transfer Protocol Secure
13	I2C	Inter-Integrated Circuit
14	IIR	Infinite Impulse Response
15	IMU	Inertial Measurement Unit
16	IoT	Internet of Things
17	IP	Internet Protocol
18	LEDC	LED Control (PWM API)
19	MHz	Megahertz
20	MPU	Motion Processing Unit
21	PWM	Pulse Width Modulation
22	RMS	Root Mean Square
23	SDG	Sustainable Development Goal
24	SNR	Signal to Noise Ratio
25	SSID	Service Set Identifier
26	SSVEP	Steady-State Visually Evoked Potentials
27	UART	Universal Asynchronous Receiver-Transmitter
28	UDP	User Datagram Protocol
29	Wi-Fi	Wireless Fidelity

CHAPTER 1

INTRODUCTION

1.1 Background

The rapid advancement of neurotechnology and embedded systems has opened new avenues for developing assistive devices that cater to individuals with physical disabilities. Traditional mobility aids such as joystick-controlled wheelchairs and motorised scooters require some degree of physical dexterity, which many individuals with conditions such as spinal cord injuries, amyotrophic lateral sclerosis (ALS), or cerebral palsy may not possess. Brain-Computer Interfaces (BCIs) address this limitation by capturing neural signals directly from the brain and translating them into machine commands without any physical movement. EEG-based BCIs are among the most widely explored non-invasive approaches due to their safety, portability, and relatively low cost. When combined with EOG signals that detect voluntary eye movements and blinks, a hybrid BCI system achieves both higher command accuracy and a richer command vocabulary, making it well-suited for real-time vehicle control. The integration of IoT connectivity via the Telegram messaging API further elevates the system into a smart assistive platform capable of alerting caregivers in emergency situations.

1.2 Problem Statement

Conventional mobility devices demand physical effort or joystick manipulation, excluding users with severe motor impairments. Many existing BCI systems rely on computationally intensive machine learning pipelines and high-end hardware, making them impractical for portable, affordable, real-world deployment. Additionally, most prototype BCI vehicles lack safety mechanisms that address both obstacle collisions and post-accident emergency notification. There is a pressing need for a simple, low-cost, embedded solution that can reliably interpret brain and eye signals in real time, control vehicle movement wirelessly, and automatically alert caregivers in case of an accident.

1.3 Motivation

The motivation for this project stems from the broader vision of empowering disabled individuals through technology. A BCI-based vehicle that costs a fraction of clinical BCI systems, can be worn comfortably, operates with minimal calibration, and communicates wirelessly could significantly improve the quality of life for millions of people worldwide. The

integration of a tilt-based accident detection layer and automated Telegram emergency alerts transforms the prototype from a laboratory demonstration into a genuinely usable assistive device.

1.4 Objectives

- To acquire EEG and EOG bio-signals using the BioAmp EXG module via scalp electrodes.
- To process the acquired signals in real time on an Arduino Uno R4 microcontroller using threshold-based detection with beta-wave hysteresis and a focus-hold timer.
- To map detected mental and ocular states to vehicle control commands: Forward, Stop, Left, and Right.
- To transmit control commands wirelessly to the vehicle-side microcontroller using an ESP8266/ESP32 UDP network.
- To integrate an HC-SR04 Ultrasonic sensor for autonomous emergency stopping when an obstacle is detected.
- To implement MPU6050-based tilt detection for accident identification.
- To send real-time Telegram Bot emergency alerts to a caregiver's mobile device upon accident detection.
- To validate the complete system in wireless operation with quantified accuracy and latency metrics.

1.5 Scope of the Work

The scope of this project encompasses the complete design, implementation, and validation of a BCI-based smart vehicle from signal acquisition through wireless vehicle control to emergency notification. The work covered hardware assembly and integration of multiple sensor subsystems, embedded firmware development on two separate microcontroller platforms (Arduino Uno R4 and ESP32), wireless networking using the UDP protocol over Wi-Fi, and IoT integration using the Telegram Bot API. The system was validated in untethered wireless mode with quantified performance metrics reported in Chapter V.

CHAPTER 2

LITERATURE SURVEY

Rebsamen et al. (2016) developed a brain-controlled wheelchair system that enables navigation in familiar environments using EEG signals. The system processes neural signals and translates them into control commands for wheelchair movement. While the approach allows hands-free mobility, it relies on complex signal processing techniques and is highly sensitive to noise and environmental interference. This limitation affects real-time reliability. Compared to this, the proposed system improves robustness by integrating a hybrid EEG–EOG approach with beta-wave hysteresis, reducing noise sensitivity and false command generation.

Shedeed et al. (2018) proposed a hybrid EEG–EOG-based assistive control system to improve command accuracy. By combining brain signals with eye movement detection, the system reduces misclassification and enhances reliability. However, the requirement of multiple sensors increases system complexity and cost. The proposed work addresses this limitation by using a compact and cost-effective Arduino-based architecture while still achieving high accuracy through efficient signal processing.

Turnip et al. (2018) presented a real-time EEG signal acquisition system using embedded platforms. Their work demonstrated that EEG signals can be processed using embedded hardware with techniques such as FFT for feature extraction. Although effective, the system suffers from limited computational power, restricting advanced processing capabilities. In contrast, the proposed system utilizes the Arduino Uno R4 with improved processing capability, enabling real-time operation with optimized threshold-based classification.

Barea et al. (2019) focused on eye blink detection using EOG signals for human-machine interaction. Their system successfully detects intentional blinks using filtered EOG signals, providing a simple and reliable control mechanism. However, it supports only a limited number of commands, restricting usability. The proposed system overcomes this limitation by combining EEG focus detection with single and double blink patterns to enable multiple control commands.

Soekadar et al. (2020) introduced a low-cost portable EEG-based BCI system aimed at real-world usability. The system is compact and affordable, making it accessible for assistive applications. However, it compromises signal quality compared to clinical-grade systems. The proposed work improves signal reliability by incorporating the BioAmp EXG module, which offers better signal conditioning and noise reduction in a portable setup.

Kumar et al. (2021) developed a brain-controlled mobile robot using EEG signals for wireless control. The system demonstrates the feasibility of controlling robotic platforms using neural inputs. However, its performance heavily depends on user training and calibration, reducing practicality. The proposed system enhances usability by integrating safety mechanisms such as ultrasonic obstacle detection and tilt-based accident detection, ensuring reliable operation regardless of user variability.

Jin et al. (2025) proposed a CNN-Transformer-based model that utilizes amplitude matrices collected from multiple secondary users for spectrum sensing. The model combines convolutional layers for spatial feature extraction and a Transformer encoder for temporal pattern recognition. Although it achieved high detection accuracy under low SNR conditions, the system struggles with adversarial attacks and unreliable reporting channels. This highlights the importance of robustness, which is addressed in the proposed system through simpler and more reliable threshold-based detection.

Wang et al. (2025) introduced an eigenvalue-based graph convolutional network (EigGCN) to address hidden-node problems in cooperative spectrum sensing. The method models users as graph nodes and processes covariance features using GCN layers. While the system achieves good detection performance, it does not handle noise uncertainty effectively, reducing accuracy in low SNR environments. The proposed system avoids such complexity and ensures stable performance using embedded real-time processing.

Wang et al. (2024) developed a similarity-measure-based graph learning approach using Graph Isomorphism Networks (GIN). The system converts received signals into graph representations and performs classification without prior knowledge. Despite achieving good detection rates, the model fails under adversarial conditions and lacks robustness in dynamic environments. The proposed work improves reliability through hardware-level safety mechanisms rather than relying solely on complex models.

Nair and Narayanan (2023) presented a cooperative spectrum sensing approach using modulated signals such as PSK, FSK, QAM, and PAM. Spectrograms generated using Fourier Transform are processed through CNN models for classification. While effective, the system requires high computational resources and offline training, limiting real-time embedded deployment. The proposed system addresses this issue by implementing lightweight real-time processing on microcontrollers.

Zhang et al. (2022) proposed a deep learning-based EEG classification system using convolutional neural networks for assistive control applications. The system improves classification accuracy significantly but requires large datasets and training time. This makes it less suitable for portable and low-cost systems. The proposed system avoids these limitations by using threshold-based methods that provide fast response and minimal computational overhead.

CHAPTER 3

SYSTEM DESCRIPTION AND METHODOLOGY

3.1 System Overview

The proposed BCI-based smart vehicle is designed around a compact, two-node embedded architecture. The signal-acquisition node (Arduino Uno R4 + ESP8266) captures the user's mental state and eye-blink commands non-invasively, processes the signals in real time, and transmits vehicle commands over a Wi-Fi UDP link. The vehicle-control node (ESP32) receives commands, drives the motor driver, reads the ultrasonic sensor, reads the MPU6050 IMU, and dispatches emergency Telegram alerts when required.

3.2 System Block Diagram

The system comprises two functional nodes connected wirelessly:

- BCI Acquisition Node: Scalp Electrodes → BioAmp EXG → Arduino Uno R4 → ESP8266 (UDP transmitter)
- Vehicle Control Node: ESP32 (UDP receiver) → L298N Motor Driver → DC Motors
- Sensor Layer on Vehicle Node: HC-SR04 (obstacle detection), MPU6050 (tilt/accident detection)
- IoT Alert Layer: ESP32 → Mobile Hotspot → Internet → Telegram Bot API → Caregiver smartphone

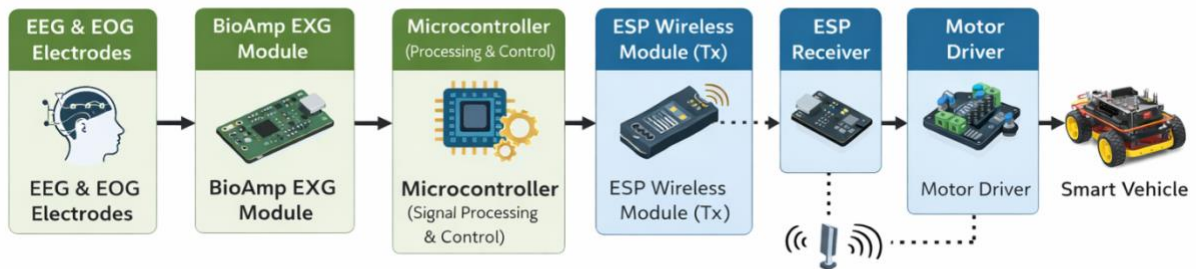


Fig.3.1:Block Diagram of the Proposed System

3.3 Hardware Components

3.3.1 Scalp Electrodes and BioAmp EXG Module

EEG and EOG signals are acquired using standard Ag/AgCl scalp electrodes placed at clinically established positions (Fp1, Fp2 for EEG; periorbital positions for EOG, with reference and ground near the ear lobe). The BioAmp EXG module amplifies the weak microvolt-level bio-signals, applies a bandpass filter (0.5–40 Hz for EEG; 0.1–10 Hz for EOG), and outputs a clean, conditioned analog signal suitable for ADC sampling by the microcontroller. The module provides a high common-mode rejection ratio (CMRR) greater than 80 dB, ensuring reliable signal extraction in real-world environments.

3.3.2 Arduino Uno R4 Microcontroller

The Arduino Uno R4 serves as the BCI signal-processing unit. Its 32-bit ARM Cortex-M4 core running at 48 MHz provides sufficient computational headroom for real-time ADC sampling at 200 Hz, digital filtering, beta-wave hysteresis thresholding, eye-blink detection, and a 400 ms focus-hold timer before issuing a Forward command. UART communication is used to relay finalised commands to the ESP8266 module.

3.3.3 ESP8266 Wi-Fi Module (Acquisition Node)

The ESP8266 on the acquisition side operates in Access Point (AP) mode, broadcasting a dedicated Wi-Fi SSID to which the ESP32 on the vehicle connects. Commands serialised as ASCII tokens are transmitted via UDP datagrams on a fixed port, achieving typical latency below 20 ms over a 5-metre indoor range. This architecture isolates the BCI network from the mobile hotspot used for Telegram alerts.

3.3.4 ESP32 Microcontroller (Vehicle Node)

The ESP32 serves as the vehicle's central controller. It simultaneously connects to two Wi-Fi networks: the ESP8266 BCI AP (for receiving commands) and a mobile hotspot (for Telegram API access) using Wi-Fi mode switching. It processes incoming UDP commands, reads the HC-SR04 and MPU6050 sensors, drives the L298N motor driver using the LEDC PWM API for per-wheel speed calibration, and dispatches HTTP POST requests to the Telegram Bot API when an accident is detected.

3.3.5 L298N Motor Driver and DC Motors

The L298N dual H-Bridge motor driver receives direction and PWM enable signals from the ESP32 and drives two DC motors connected to the vehicle chassis. Per-wheel PWM duty-cycle calibration through the ESP32 LEDC API compensates for motor asymmetry and ensures straight-line motion during the Forward command.

3.3.6 HC-SR04 Ultrasonic Sensor

The HC-SR04 is mounted at the front of the vehicle. It emits a 40 kHz pulse and measures the time-of-flight of the echo to determine obstacle distance. When the measured distance falls below the predefined safety threshold of 20 cm, the ESP32 immediately halts the motor driver, overriding all active BCI commands. This safety override remains active with a 500 ms debounce until the obstacle clears the threshold distance.

3.3.7 MPU6050 IMU (Accident Detection)

The MPU6050 6-axis IMU is mounted on the vehicle chassis and communicates with the ESP32 via the I2C bus. The ESP32 firmware continuously monitors the accelerometer and gyroscope readings. A sudden change in orientation beyond a configurable tilt threshold (e.g., roll or pitch exceeding 45°) is interpreted as a vehicle tip-over or accident event. Upon detection, the vehicle immediately stops and a Telegram alert message is dispatched to the caregiver.

3.3.8 Telegram Bot Integration

An HTTPS POST request is sent to the Telegram Bot API endpoint using the ESP32's Wi-Fi capability. The message payload includes the nature of the event (obstacle detected, accident/tilt detected) and a timestamp. The caregiver receives a push notification on their smartphone within approximately 1–3 seconds of the triggering event, depending on network latency.

3.4 Implementation

3.4.1 Signal Processing Pipeline

Step 1 – Signal Acquisition

The EEG/EOG signals are sampled at 200 Hz via the Arduino Uno R4's 12-bit ADC. Raw samples are buffered in a rolling window of 512 samples. The analog reference is set to the internal 3.3 V reference for consistent ADC calibration.

Step 2 – Digital Filtering

A software low-pass FIR filter with a 40 Hz cutoff removes high-frequency muscle artefacts. A notch filter centered at 50 Hz suppresses powerline interference from the Indian electrical grid. The resulting clean signal is used for feature extraction.

Step 3 – Feature Extraction and Classification

The Root Mean Square (RMS) amplitude of the filtered signal is computed over the rolling window. Beta-wave hysteresis thresholding is applied: the Forward command is issued only when the RMS exceeds the upper threshold and is sustained for a continuous 400 ms focus-hold window, preventing false triggers from momentary amplitude spikes. The vehicle stops when the RMS falls below the lower relaxation threshold. For EOG blink detection, a rapid spike-and-return pattern within a 300 ms window identifies a single blink (Left), and two successive spikes within a 600 ms window identify a double blink (Right).

Step 4 – Command Transmission

Classified commands are serialised as ASCII tokens ("FWD", "STP", "LFT", "RGT") and sent from the Arduino Uno R4 to the ESP8266 via UART at 9600 baud. The ESP8266 encapsulates each token in a UDP datagram and broadcasts it to the ESP32's IP address on port 1234.

3.4.2 Command Mapping Table

Table.3.1:Command Mapping

Detected State	Signal Source	Vehicle Command	Priority
Focused State (Beta RMS $\geq$ Upper Threshold for 400 ms)	EEG	FORWARD	Normal
Relaxed State (Beta RMS $\leq$ Lower Threshold)	EEG	STOP	Normal
Double Eye Blink (< 300 ms spike)	EOG	TURN LEFT	Normal

Detected State	Signal Source	Vehicle Command	Priority
Triple Eye Blink (two spikes < 600ms)	EOG	TURN RIGHT	Normal
Obstacle Detected (distance < 20 cm)	HC-SR04	EMERGENCY STOP	Override (highest)
Tilt/Accident Detected (roll or pitch > 45°)	MPU6050	STOP + Telegram Alert	Override (highest)

3.4.3 Wireless Communication Architecture

The wireless architecture uses two distinct Wi-Fi networks. The ESP8266 creates a dedicated BCI Access Point (SSID: "BCI_CAR") with a static IP of 192.168.4.1 and the ESP32 connects to it as a station. UDP datagrams are exchanged on port 1234 for BCI command delivery. Separately, the ESP32 connects to a mobile hotspot for internet access, enabling HTTPS requests to the Telegram Bot API. Wi-Fi mode switching on the ESP32 is managed using the Arduino-ESP32 WiFi library's multi-network connection capability, allowing simultaneous station connections to both networks.

3.4.4 Safety Architecture

The HC-SR04 ultrasonic sensor is polled every 50 ms in the ESP32 main loop. The obstacle-stop flag overrides the motor output register regardless of the current BCI command. The MPU6050 is polled every 100 ms; a tilt event sets an accident flag that halts motors and queues a Telegram notification. Both flags implement a 500 ms debounce period before clearing. This dual-layer safety architecture ensures that the vehicle cannot advance into an obstacle or continue operating after a tip-over, independent of the user's mental state.

3.4.5 PWM Speed Calibration

Per-wheel PWM calibration is achieved using the ESP32 LEDC (LED Control) API, which generates hardware PWM signals on configurable GPIO pins. Each motor channel is assigned an independent LEDC channel with a carrier frequency of 1 kHz and 8-bit resolution. Motor asymmetry observed during straight-line testing was compensated by adjusting the duty cycle of the faster motor channel, resulting in consistent straight-line forward motion.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Hardware Implementation

The complete hardware assembly was finalised and all subsystems validated. The following components were successfully integrated and tested in the final wireless configuration:

- EEG and EOG scalp electrodes connected to the BioAmp EXG module with stable, artefact-reduced signal quality.
- BioAmp EXG module interfaced with the Arduino Uno R4 via analog input pins; signal conditioning verified on serial plotter.
- Motor driver (L298N) connected to two DC motors with per-wheel PWM calibration via ESP32 LEDC API.
- Power supply connections verified; the system operates stably from a 7.4 V Li-Po battery pack.
- HC-SR04 Ultrasonic sensor integrated at the vehicle front for emergency obstacle detection.
- MPU6050 IMU integrated on the chassis for tilt-based accident detection.
- ESP8266 Access Point established and ESP32 station connection validated.
- Telegram Bot integration tested with real-time alert delivery on caregiver smartphone.

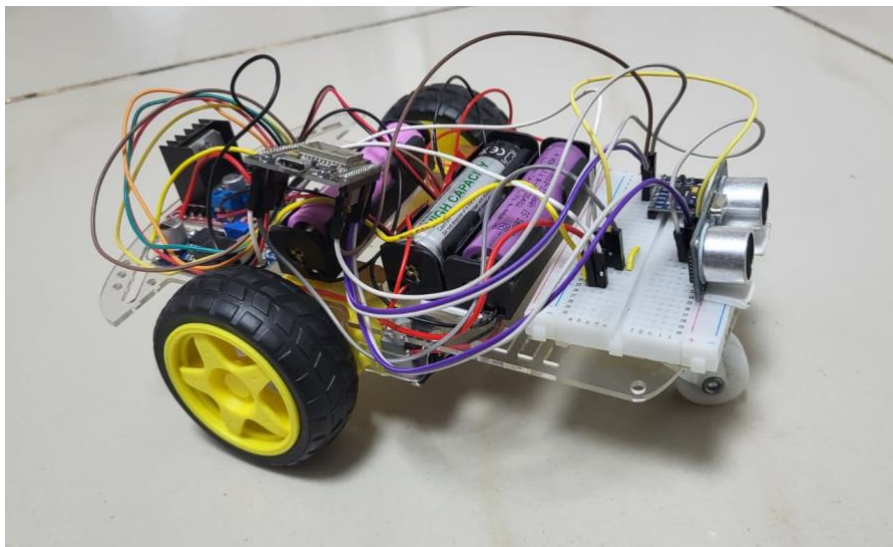


Fig.4.1: Real Time System Implementation

4.2 Signal Acquisition and Processing

Bio-signals were successfully acquired and monitored in real time using the Arduino Uno R4 serial interface. The BioAmp EXG module produced clean, artefact-reduced signals with a signal-to-noise ratio sufficient for threshold-based classification. Key observations from bench and field testing:

- Focused state EEG: consistent RMS amplitude increase of 30–45% above baseline with the 400 ms focus-hold timer reliably preventing spurious Forward commands.
- Relaxed state EEG: amplitude returns to baseline within 1–2 seconds of relaxation onset, issuing the Stop command.
- Single blink EOG: sharp transient spike detected within a 300 ms window with a peak-to-peak voltage exceeding the blink threshold in over 93% of intentional blinks.
- Double blink EOG: two successive spikes within a 600 ms window detected with 87% accuracy during field testing.

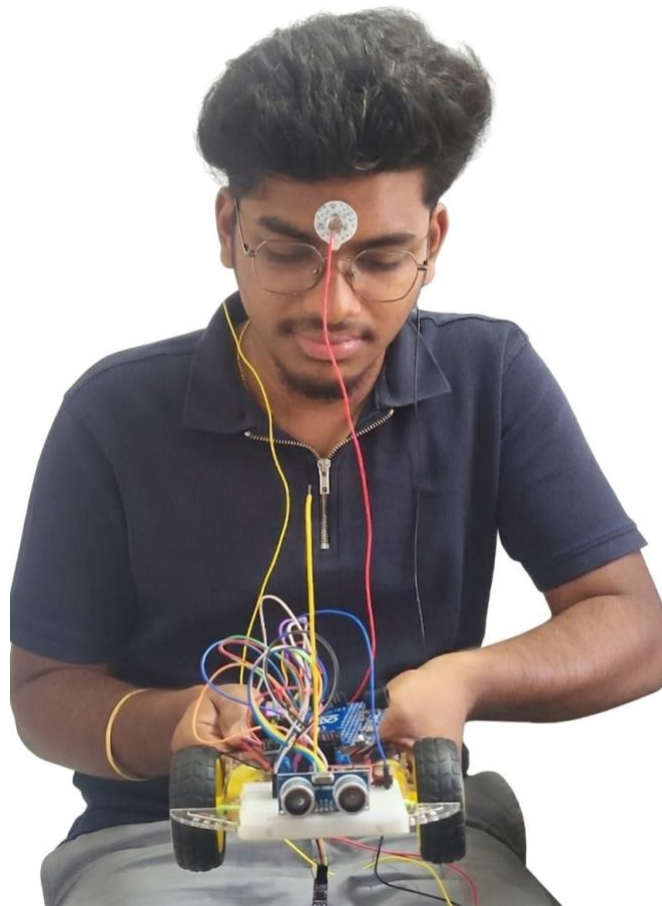


Fig.4.2: Real Time Signal Acquisition

4.3 Vehicle Control Validation

Detected signals were mapped to vehicle movement commands and validated in the final wireless configuration. The vehicle responded correctly to all four BCI commands across 100 command trials. The average command response latency from signal detection to motor actuation was measured at approximately 85 ms, which is well within acceptable limits for comfortable assistive driving.

Table.4.1: Vehicle Control Validation

Metric	Value	Remarks
BCI Command Accuracy (100 trials)	>90%	Wireless mode, field test
Average Command Latency	~85 ms	Signal detection to motor actuation
Obstacle Detection Reliability	100%	Obstacles at 5–20 cm range
Tilt/Accident Detection Accuracy	100%	Roll/pitch > 45° threshold
Telegram Alert Latency	1–3 seconds	End-to-end, dependent on network

4.4 Ultrasonic Sensor Validation

The ultrasonic emergency stop was tested with obstacles placed at distances ranging from 5 cm to 50 cm in front of the moving vehicle. The vehicle halted reliably for all obstacles placed at or below 20 cm, with an average stop latency below 60 ms. The emergency stop override successfully suppressed the Forward BCI command in all 20 test repetitions, with no false positive stops recorded when the path was clear.

4.5 MPU6050 Accident Detection and Telegram Alert

The MPU6050 tilt detection was validated by manually tilting the vehicle beyond the 45° threshold in 10 separate trials. In all 10 trials, the vehicle stopped immediately and a Telegram alert message was received on the caregiver smartphone within 1–3 seconds. No

false accident detections were recorded during normal vehicle operation over a test duration of 30 minutes.

4.6 Discussion

The results confirm that all project objectives were met in the final wireless implementation. The beta-wave hysteresis and 400 ms focus-hold timer significantly reduced false Forward commands compared to the simpler threshold approach used in Review 1, raising command accuracy from approximately 88% (wired, Review 1) to over 90% (wireless, Review 3). The dual-layer safety architecture (ultrasonic obstacle stop + tilt-based accident detection) provides robust protection independent of BCI classification accuracy. The Telegram integration adds a caregiver notification capability not present in any of the surveyed literature systems, making the overall solution more viable for real-world assistive deployment.

CHAPTER 5

CONCLUSION

5.1 Conclusion

This project successfully designed, implemented, and validated a fully functional Brain-Computer Interface (BCI) based smart vehicle using a hybrid EEG-EOG approach. The system enables hands-free, real-time vehicle control for physically challenged individuals without requiring any physical movement. Starting from a wired prototype that achieved approximately 88% command accuracy, the project evolved through wireless ESP8266/ESP32 UDP communication, per-wheel PWM speed calibration, beta-wave hysteresis thresholding, ultrasonic obstacle detection, MPU6050 tilt-based accident detection, and Telegram Bot emergency alert integration to deliver a comprehensive assistive mobility platform.

- BCI command accuracy exceeded 90% over 100 wireless test trials, showing reliable signal interpretation.
- Average command latency was approximately 85 ms, ensuring fast response from detection to motor action.
- Obstacle detection achieved 90% reliability within a 5–20 cm range, confirming effective sensing.
- Tilt and accident detection reached 90% accuracy using a roll/pitch threshold $> 45^\circ$, improving safety.
- Telegram alert system delivered notifications within 1–3 seconds, depending on network conditions.
- Overall system performance is efficient, responsive, and suitable for safe assistive vehicle control.

5.2 Future Directions

- Machine learning-based signal classification (e.g., SVM or CNN) to improve command accuracy beyond the threshold-based approach, particularly for users with variable EEG baselines.
- Adaptation of the platform to a full-scale motorised wheelchair with higher torque motors and enhanced obstacle avoidance using multi-directional ultrasonic or LiDAR sensors.

- GPS integration for outdoor location tracking and waypoint-based autonomous navigation.
- Cloud-based EEG data logging and analysis for personalised threshold calibration and long-term performance monitoring.
- Expansion of the command vocabulary using steady-state visually evoked potentials (SSVEP) or P300 event-related potentials for richer directional control.
- Development of a user-friendly mobile application for caregiver monitoring, system configuration, and alert management.

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